# PART II : CHEMISTRY

**SECTION 1 (Maximum Marks: 28)**

- This section contains **SEVEN** questions
- Each question has **FOUR** options [A], [B], [C] and [D]. **ONE OR MORE THAN ONE** of these four options is(are) correct
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- For each question, marks will be awarded in one of the following categories:

| | | |
|---|---|---|
| *Full Marks* | : +4 | If only the bubble(s) corresponding to all the correct option(s) is(are) darkened |
| *Partial Marks* | : +1 | For darkening a bubble corresponding **to each correct option**, provided NO incorrect option is darkened |
| *Zero Marks* | : 0 | If none of the bubbles is darkened |
| *Negative Marks* | : -2 | In all other cases |

- For example, if [A], [C] and [D] are all the correct options for a question, darkening all these three will get +4 marks; darkening only [A] and [D] will get +2 marks; and darkening [A] and [B] will get -2 marks, as a wrong option is also darkened

Q.19 An ideal gas is expanded from ($p_1$, $V_1$, $T_1$) to ($p_2$, $V_2$, $T_2$) under different conditions. The correct statement(s) among the following is(are)

[A] The work done on the gas is maximum when it is compressed irreversibly from ($p_2$, $V_2$) to ($p_1$, $V_1$) against constant pressure $p_1$

[B] If the expansion is carried out freely, it is simultaneously both isothermal as well as adiabatic

[C] The work done by the gas is less when it is expanded reversibly from $V_1$ to $V_2$ under adiabatic conditions as compared to that when expanded reversibly from $V_1$ to $V_2$ under isothermal conditions

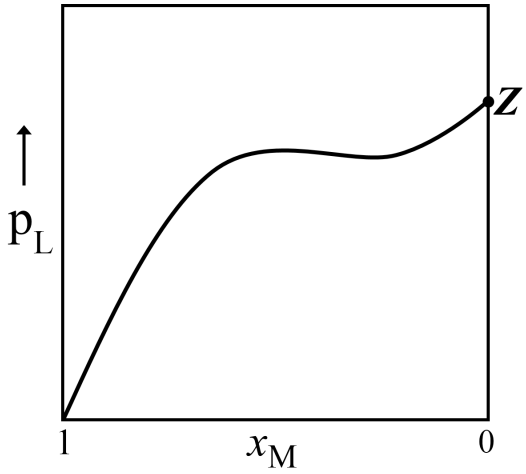
[D] The change in internal energy of the gas is (i) zero, if it is expanded reversibly with $T_1 = T_2$, and (ii) positive, if it is expanded reversibly under adiabatic conditions with $T_1 \neq T_2$

**Space for rough work**

**Answer for the above question**

**Ans for Q.19: (A), (B), and (C)**

Q.20 For a solution formed by mixing liquids **L** and **M**, the vapour pressure of **L** plotted against the mole fraction of **M** in solution is shown in the following figure. Here $x_L$ and $x_M$ represent mole fractions of **L** and **M**, respectively, in the solution. The correct statement(s) applicable to this system is(are)




[A] The point ***Z*** represents vapour pressure of pure liquid **M** and Raoult's law is obeyed from $x_L = 0$ to $x_L = 1$

[B] The point ***Z*** represents vapour pressure of pure liquid **L** and Raoult's law is obeyed when $x_L \rightarrow 1$

[C] The point ***Z*** represents vapour pressure of pure liquid **M** and Raoult's law is obeyed when $x_L \rightarrow 0$

[D] Attractive intermolecular interactions between **L**-**L** in pure liquid **L** and **M**-**M** in pure liquid **M** are stronger than those between **L**-**M** when mixed in solution

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**Space for rough work**

## Answer for the above question

**Ans for Q.20: (B) and (D)**

Q.21 The correct statement(s) about the oxoacids, $HClO_4$ and HClO, is(are)

[A] The central atom in both $HClO_4$ and HClO is $sp^3$ hybridized

[B] $HClO_4$ is more acidic than HClO because of the resonance stabilization of its anion

[C] $HClO_4$ is formed in the reaction between $Cl_2$ and $H_2O$

[D] The conjugate base of $HClO_4$ is weaker base than $H_2O$

Q.22 The colour of the $X_2$ molecules of group 17 elements changes gradually from yellow to violet down the group. This is due to

[A] the physical state of $X_2$ at room temperature changes from gas to solid down the group

[B] decrease in ionization energy down the group

[C] decrease in $\pi^*$-$\sigma^*$ gap down the group

[D] decrease in HOMO-LUMO gap down the group

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**Space for rough work**

## Answers for the above questions

| Ans for Q.21: (A), (B), and (D) | Ans for Q.22: (C) and (D) |
|---|---|

Q.23 Addition of excess aqueous ammonia to a pink coloured aqueous solution of $MCl_2 \cdot 6H_2O$ (**X**) and $NH_4Cl$ gives an octahedral complex **Y** in the presence of air. In aqueous solution, complex **Y** behaves as 1:3 electrolyte. The reaction of **X** with excess HCl at room temperature results in the formation of a blue coloured complex **Z**. The calculated spin only magnetic moment of **X** and **Z** is 3.87 B.M., whereas it is zero for complex **Y**.
Among the following options, which statement(s) is(are) correct?

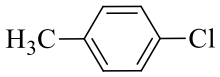
[A] Addition of silver nitrate to **Y** gives only two equivalents of silver chloride

[B] The hybridization of the central metal ion in **Y** is $d^2sp^3$

[C] **Z** is a tetrahedral complex

[D] When **X** and **Z** are in equilibrium at 0°C, the colour of the solution is pink

Q.24 The IUPAC name(s) of the following compound is(are)




[A] 1-chloro-4-methylbenzene

[B] 4-chlorotoluene

[C] 4-methylchlorobenzene

[D] 1-methyl-4-chlorobenzene

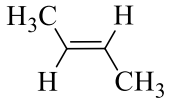
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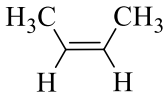
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## Answers for the above questions

| Ans for Q.23: (B), (C), and (D) | Ans for Q.24: (A) and (B) |
|---|---|

Q.25 The correct statement(s) for the following addition reactions is(are)

(i)  $\xrightarrow{Br_2 / CHCl_3}$ **M** and **N**

(ii)  $\xrightarrow{Br_2 / CHCl_3}$ **O** and **P**

[A] **O** and **P** are identical molecules

[B] (**M** and **O**) and (**N** and **P**) are two pairs of diastereomers

[C] (**M** and **O**) and (**N** and **P**) are two pairs of enantiomers

[D] Bromination proceeds through *trans*-addition in both the reactions

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**Space for rough work**

## Answer for the above question

**Ans for Q.25: (B) and (D)**

**SECTION 2 (Maximum Marks: 15)**

- This section contains **FIVE** questions
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive
- For each question, darken the bubble corresponding to the correct integer in the ORS
- For each question, marks will be awarded in one of the following categories:

*Full Marks* : +3 If only the bubble corresponding to the correct answer is darkened
*Zero Marks* : 0 In all other cases

Q.26 A crystalline solid of a pure substance has a face-centred cubic structure with a cell edge of 400 pm. If the density of the substance in the crystal is 8 g $cm^{-3}$, then the number of atoms present in 256 g of the crystal is $N \times 10^{24}$. The value of $N$ is

Q.27 The conductance of a 0.0015 M aqueous solution of a weak monobasic acid was determined by using a conductivity cell consisting of platinized Pt electrodes. The distance between the electrodes is 120 cm with an area of cross section of 1 $cm^2$. The conductance of this solution was found to be $5 \times 10^{-7}$ S. The pH of the solution is 4. The value of limiting molar conductivity $\left(\Lambda_m^o\right)$ of this weak monobasic acid in aqueous solution is $Z \times 10^2$ S $cm^{-1}$ $mol^{-1}$. The value of $Z$ is

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**Space for rough work**

## Answers for the above questions

| Ans for Q.26: (2) | Ans for Q.27: (6) |
|---|---|

Q.28 The sum of the number of lone pairs of electrons on each central atom in the following species is

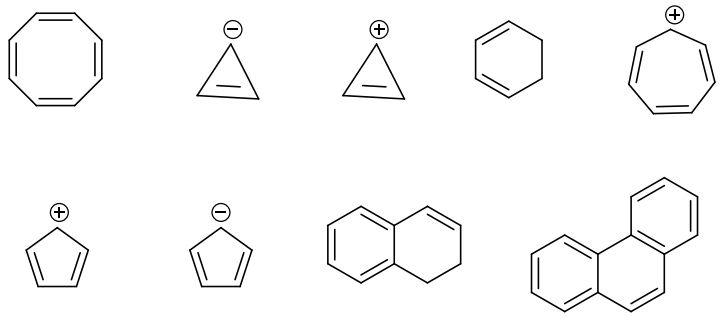
$[TeBr_6]^{2-}$, $[BrF_2]^+$, $SNF_3$, and $[XeF_3]^-$

(Atomic numbers: N = 7, F = 9, S = 16, Br = 35, Te = 52, Xe = 54)

Q.29 Among $H_2$, $He_2^+$, $Li_2$, $Be_2$, $B_2$, $C_2$, $N_2$, $O_2^-$, and $F_2$, the number of diamagnetic species is

(Atomic numbers: H = 1, He = 2, Li = 3, Be = 4, B = 5, C = 6, N = 7, O = 8, F = 9)

Q.30 Among the following, the number of aromatic compound(s) is



**Space for rough work**

## Answers for the above questions

**Ans for Q.28: (6)**

**Ans for Q.30: (5)**

**Due to printing inconsistency in Q.29 in Code1 of Paper1 (and the corresponding question in all other codes), +3 marks will be awarded to all candidates for this question.**

**SECTION 3 (Maximum Marks: 18)**

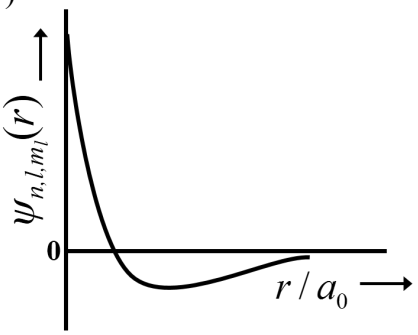
- This section contains **SIX** questions of matching type
- This section contains **TWO** tables (each having 3 columns and 4 rows)
- Based on each table, there are **THREE** questions
- Each question has **FOUR** options [A], [B], [C], and [D]. **ONLY ONE** of these four options is correct
- For each question, darken the bubble corresponding to the correct option in the ORS
- For each question, marks will be awarded in one of the following categories:

| | | |
|---|---|---|
| *Full Marks* | : +3 | If only the bubble corresponding to the correct option is darkened |
| *Zero Marks* | : 0 | If none of the bubbles is darkened |
| *Negative Marks*: | -1 | In all other cases |

**Space for rough work**

**Answer Q.31, Q.32 and Q.33 by appropriately matching the information given in the three columns of the following table.**

The wave function, $\psi_{n,l,m_l}$ is a mathematical function whose value depends upon spherical polar coordinates $(r, \theta, \phi)$ of the electron and characterized by the quantum numbers $n$, $l$ and $m_l$. Here $r$ is distance from nucleus, $\theta$ is colatitude and $\phi$ is azimuth. In the mathematical functions given in the Table, $Z$ is atomic number and $a_o$ is Bohr radius.

| Column 1 | Column 2 | Column 3 |
|---|---|---|
| (I) 1s orbital | (i) $\psi_{n,l,m_l} \propto \left(\frac{Z}{a_o}\right)^{\frac{3}{2}} e^{-\left(\frac{Zr}{a_o}\right)}$ | (P)   |
| (II) 2s orbital | (ii) One radial node | (Q) Probability density at nucleus $\propto \frac{1}{a_o^3}$ |
| (III) $2p_z$ orbital | (iii) $\psi_{n,l,m_l} \propto \left(\frac{Z}{a_o}\right)^{\frac{5}{2}} re^{-\left(\frac{Zr}{2a_o}\right)} \cos\theta$ | (R) Probability density is maximum at nucleus |
| (IV) $3d_z^2$ orbital | (iv) *xy*-plane is a nodal plane | (S) Energy needed to excite electron from $n = 2$ state to $n = 4$ state is $\frac{27}{32}$ times the energy needed to excite electron from $n = 2$ state to $n = 6$ state |

Q.31 For the given orbital in Column 1, the only CORRECT combination for any hydrogen-like species is

[A] (I) (ii) (S) [B] (IV) (iv) (R) [C] (II) (ii) (P) [D] (III) (iii) (P)

Q.32 For hydrogen atom, the only CORRECT combination is

[A] (I) (i) (S) [B] (II) (i) (Q) [C] (I) (i) (P) [D] (I) (iv) (R)

Q.33 For $He^+$ ion, the only **INCORRECT** combination is

[A] (I) (i) (R) [B] (II) (ii) (Q) [C] (I) (iii) (R) [D] (I) (i) (S)

**Space for rough work**

## Answers for the above questions

| Ans for Q.31: (C) | Ans for Q.32: (A) | Ans for Q.33: (C) |
|---|---|---|

**Answer Q.34, Q.35 and Q.36 by appropriately matching the information given in the three columns of the following table.**

| Columns 1, 2 and 3 contain starting materials, reaction conditions, and type of reactions, respectively. | | |
|---|---|---|
| **Column 1** | **Column 2** | **Column 3** |
| (I) Toluene | (i) NaOH/ $Br_2$ | (P) Condensation |
| (II) Acetophenone | (ii) $Br_2$/ h$\nu$ | (Q) Carboxylation |
| (III) Benzaldehyde | (iii) $(CH_3CO)_2O$/ $CH_3COOK$ | (R) Substitution |
| (IV) Phenol | (iv) NaOH/ $CO_2$ | (S) Haloform |

Q.34 For the synthesis of benzoic acid, the only CORRECT combination is

[A] (II) (i) (S) [B] (IV) (ii) (P) [C] (I) (iv) (Q) [D] (III) (iv) (R)

Q.35 The only CORRECT combination that gives two different carboxylic acids is

[A] (II) (iv) (R) [B] (IV) (iii) (Q) [C] (III) (iii) (P) [D] (I) (i) (S)

Q.36 The only CORRECT combination in which the reaction proceeds through radical mechanism is

[A] (III) (ii) (P) [B] (IV) (i) (Q) [C] (II) (iii) (R) [D] (I) (ii) (R)

**END OF PART II : CHEMISTRY**

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## Answers for the above questions

| Ans for Q.34: (A) | Ans for Q.35: (C) | Ans for Q.36: (D) |
|---|---|---|